

Priyanga B

📍 Kerala, India | ✉️ priyangagbr@gmail.com | [in priyangahere](#) | [🔗 Priyanga-babu-rajendran](#) | [🌐 Portfolio](#)

SUMMARY

Data Science postgraduate researcher focused on the convergence of artificial intelligence, quantum computing, and cybersecurity. Experienced in developing machine learning pipelines, distributed analytics, and hybrid AI systems. Currently conducting research on privacy-preserving federated learning, quantum machine learning, and trustworthy AI, with an emphasis on privacy, security, robustness, and rigorous model evaluation.

EDUCATION

M.Sc. Data Science — Vellore Institute of Technology, Chennai <i>CGPA: 8.34/10</i>	2025–2027
B.Sc. Physics — Sree Krishna College, Guruvayur, Kerala <i>CGPA: 7.967/10</i>	2021–2024
Higher Secondary — Biology Science with Maths — Kerala State Board <i>Percentage: 99%</i>	2019–2021

TECHNICAL SKILLS & EXPERTISE

Machine Learning: Deep Learning, Statistical Modeling, Natural Language Processing, Model Optimization
Trustworthy & Privacy-Preserving AI: Federated Learning, Differential Privacy, Model Robustness, Model Evaluation, Privacy-Preserving Machine Learning
Programming: Python, R, SQL
Frameworks & Libraries: PyTorch, Scikit-learn, NumPy, Pandas, OpenCV, PennyLane, Qiskit
Distributed Computing: Hadoop, Spark, HDFS, MapReduce, YARN
Tools & Platforms: Jupyter Notebook, Streamlit, Git, GitHub, LaTeX, VS Code
Data Visualization: Matplotlib, Seaborn

PROJECTS

Adaptive Personalized Hybrid Quantum-Classical Federated Learning for Privacy-Preserving Intrusion Detection (*Ongoing Research*)

- Designing a personalized hybrid quantum-classical federated learning framework for privacy-preserving network intrusion detection under heterogeneous and non-IID client data.
- Integrating Differential Privacy (DP), Quantum Neural Networks (QNNs), and reliability-aware adaptive client aggregation to improve privacy, robustness, personalization, and reliability across distributed clients.

Quantum Kernel-Based Few-Shot Auditing for Real-Time Contextual-Integrity Violation Detection in LLM Agent Outputs (*Ongoing Research*)

- Developing a quantum kernel-based few-shot anomaly detection framework for real-time identification of contextual-integrity violations in LLM agent outputs.
- Designing quantum feature encodings and fidelity-based kernels with kernel SVM classification, and benchmarking against classical kernels and LLM-as-judge methods across accuracy, latency, label efficiency, and cross-benchmark generalization.

SignTalk - A Continuous Word-Level Indian Sign Language Recognition Expert System using DL and ML | [🔗 Repo](#)

- Developed a hybrid AI system integrating YOLOv5, MediaPipe-SVM, and NLP-based grammar correction for real-time Indian Sign Language recognition and gesture-to-text translation.

- Evaluated the system on a custom dataset of 82 Indian Sign Language words, achieving ~97% classification accuracy.

Distributed Web Log Analytics Using the Hadoop Ecosystem | [Repo](#)

- Built a 3-node Hadoop cluster using HDFS, YARN, MapReduce, and Spark for distributed analysis of large-scale NASA web server logs.
- Developed parallel analytics pipelines and an interactive Streamlit dashboard for scalable data processing, analysis, and visualization.

X-ray Diffraction Analysis and Cell Parameter Calculation of Reported Silicates

- Performed X-ray diffraction analysis of reported silicate materials and applied Bragg’s law and crystallographic relations to determine interplanar spacings and unit-cell parameters.
- Compared calculated crystallographic parameters with reported literature values to evaluate structural agreement.

PUBLICATIONS

Computational Methods for Multilingual Sentiment Analysis in Social Media: A Systematic Review (*Manuscript under review at IEEE Access*)

- Conducted a PRISMA-based systematic review to evaluate multilingual sentiment analysis methods across machine learning, deep learning, and transformer-based approaches.
- Analyzed reported model performance and methodological trends across multilingual and code-mixed settings, with focus on BERT and XLM-R architectures.

CERTIFICATIONS

- Python for Everybody | University of Michigan | Coursera
- Deloitte Australia – Data Analytics Job Simulation | Forage
- Introduction to Quantum Information | KAIST
- Tools for Data Science | IBM
- Agentic AI and Autonomous Intelligent Systems, 2-day Workshop | VIT, Chennai
- IQM Quantum School Certificate | IQM Quantum Computers
- Cloud Computing | NPTEL

PROFESSIONAL EXPERIENCE

Office Assistant | **Mega Motors, Hero MotoCorp** | **Aluva, Kerala** Aug 2024 – Apr 2025

- Supported daily administrative operations and maintained routine office documentation
- Assisted with insurance policy management and customer inquiries

LANGUAGES

English (Advanced) | Hindi (Advanced) | Tamil (Advanced) | Korean (Beginner) | Malayalam (Native)